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22/7/25

Program 1:

```
package com.training.ooc;
class Student {
protected String name;
protected String id;
protected int age;
protected double grade;
protected String address;
public Student() {
    this.name = "";
    this.id = "";
    this.age = 0;
    this.grade = 0.0;
    this.address = "";
}
public Student(String name, String id, int age, double grade, String address) {
    this.name = name;
    this.id = id;
    this.age = age;
    this.grade = grade;
    this.address = address;
}
public String getName() { return name; }
public void setName(String name) { this.name = name; }
public String getId() { return id; }
public void setId(String id) { this.id = id; }
public int getAge() { return age; }
public void setAge(int age) { this.age = age; }
public double getGrade() { return grade; }
public void setGrade(double grade) { this.grade = grade; }
public String getAddress() { return address; }
public void setAddress(String address) { this.address = address; }
public void display() {
    System.out.println("Name: " + name);
    System.out.println("ID: " + id);
    System.out.println("Age: " + age);
    System.out.println("Grade: " + grade);
    System.out.println("Address: " + address);
}
public boolean isPassed() {
    return grade > 50;
}
}
class UGStudent extends Student {
private String degree;
private String stream;
public UGStudent() {
```

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```
    super();
    this.degree = "";
    this.stream = "";
}
public UGStudent(String name, String id, int age, double grade, String address, String degree, String
stream) {
    super(name, id, age, grade, address);
    this.degree = degree;
    this.stream = stream;
}
public String getDegree() { return degree; }
public void setDegree(String degree) { this.degree = degree; }
public String getStream() { return stream; }
public void setStream(String stream) { this.stream = stream; }
@Override
public void display() {
    System.out.println("UG Student Details:");
    System.out.println("Name: " + name);
    System.out.println("ID: " + id);
    System.out.println("Age: " + age);
    System.out.println("Grade: " + grade);
    System.out.println("Address: " + address);
    System.out.println("Degree: " + degree);
    System.out.println("Stream: " + stream);
}
@Override
public boolean isPassed() {
    return grade > 70;
}
}
class PGStudent extends Student {
private String specialization;
private int noOfPapersPublished;
public PGStudent() {
    super();
    this.specialization = "";
    this.noOfPapersPublished = 0;
}
public PGStudent(String name, String id, int age, double grade, String address, String specialization, int
noOfPapersPublished) {
    super(name, id, age, grade, address);
    this.specialization = specialization;
    this.noOfPapersPublished = noOfPapersPublished;
}
public String getSpecialization() { return specialization; }
public void setSpecialization(String specialization) { this.specialization = specialization; }
public int getNoOfPapersPublished() { return noOfPapersPublished; }
public void setNoOfPapersPublished(int noOfPapersPublished) { this.noOfPapersPublished =
noOfPapersPublished; }
```

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```
@Override
public void display() {
    System.out.println("PG Student Details:");
    System.out.println("Name: " + name);
    System.out.println("ID: " + id);
    System.out.println("Age: " + age);
    System.out.println("Grade: " + grade);
    System.out.println("Address: " + address);
    System.out.println("Specialization: " + specialization);
    System.out.println("Number of Papers Published: " + noOfPapersPublished);
}
@Override
public boolean isPassed() {
    return grade > 70 && noOfPapersPublished >= 2;
}
}
public class main {
public static void main(String[] args) {

    Student s1 = new Student("Surya", "S100", 20, 55, "123 Elm Street");
    s1.display();
    System.out.println("Passed: " + s1.isPassed());
    System.out.println();

    UGStudent ug1 = new UGStudent("Alter", "UG101", 19, 75, "456 Oak Street", "B.Tech", "Computer
Science");
    ug1.display();
    System.out.println("Passed: " + ug1.isPassed());
    System.out.println();

    PGStudent pg1 = new PGStudent("Karthi", "PG102", 24, 80, "789 Pine Street", "Data Science", 3);
    pg1.display();
    System.out.println("Passed: " + pg1.isPassed());
    System.out.println();

}
}
```

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```
<terminated> main [Java Application] C:\U
Name: Surya
ID: S100
Age: 20
Grade: 55.0
Address: 123 Elm Street
Passed: true

UG Student Details:
Name: Alter
ID: UG101
Age: 19
Grade: 75.0
Address: 456 Oak Street
Degree: B.Tech
Stream: Computer Science
Passed: true

PG Student Details:
Name: Karthi
ID: PG102
Age: 24
Grade: 80.0
Address: 789 Pine Street
Specialization: Data Science
Number of Papers Published: 3
Passed: true
```

Program 2:

```
package com.training.ooc;
import java.util.Scanner;
class Vehicle {
protected String make;
protected String vehicleNumber;
protected String fuelType;
protected Integer fuelCapacity;
protected Integer cc;
public Vehicle(String make, String vehicleNumber, String fuelType, Integer fuelCapacity, Integer cc) {
    this.make = make;
    this.vehicleNumber = vehicleNumber;
    this.fuelType = fuelType;
    this.fuelCapacity = fuelCapacity;
    this.cc = cc;
}
public String getMake() { return make; }
public void setMake(String make) { this.make = make; }
public String getVehicleNumber() { return vehicleNumber; }
public void setVehicleNumber(String vehicleNumber) { this.vehicleNumber = vehicleNumber; }
public String getFuelType() { return fuelType; }
public void setFuelType(String fuelType) { this.fuelType = fuelType; }
public Integer getFuelCapacity() { return fuelCapacity; }
```

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```java
public void setFuelCapacity(Integer fuelCapacity) { this.fuelCapacity = fuelCapacity; }
public Integer getCc() { return cc; }
public void setCc(Integer cc) { this.cc = cc; }
public void displayMake() {
    System.out.println("***" + make + "***");
}
public void displayBasicInfo() {
    System.out.println("---Basic Information---");
    System.out.println("Vehicle Number: " + vehicleNumber);
    System.out.println("Fuel Type: " + fuelType);
    System.out.println("Fuel Capacity: " + fuelCapacity);
    System.out.println("CC: " + cc);
}
public void displayDetailInfo() {

}
}
class TwoWheeler extends Vehicle {
private Boolean kickStartAvailable;
public TwoWheeler(String make, String vehicleNumber, String fuelType, Integer fuelCapacity, Integer cc,
                Boolean kickStartAvailable) {
    super(make, vehicleNumber, fuelType, fuelCapacity, cc);
    this.kickStartAvailable = kickStartAvailable;
}
public Boolean getKickStartAvailable() { return kickStartAvailable; }
public void setKickStartAvailable(Boolean kickStartAvailable) { this.kickStartAvailable = kickStartAvailable;
}
@Override
public void displayDetailInfo() {
    System.out.println("---Detail Information---");
    System.out.println("Kick Start Available: " + (kickStartAvailable ? "Yes" : "No"));
}
}
class FourWheeler extends Vehicle {
private String audioSystem;
private Integer numberOfDoors;
public FourWheeler(String make, String vehicleNumber, String fuelType, Integer fuelCapacity, Integer cc,
                String audioSystem, Integer numberOfDoors) {
    super(make, vehicleNumber, fuelType, fuelCapacity, cc);
    this.audioSystem = audioSystem;
    this.numberOfDoors = numberOfDoors;
}
public String getAudioSystem() { return audioSystem; }
public void setAudioSystem(String audioSystem) { this.audioSystem = audioSystem; }
public Integer getNumberOfDoors() { return numberOfDoors; }
public void setNumberOfDoors(Integer numberOfDoors) { this.numberOfDoors = numberOfDoors; }
@Override
public void displayDetailInfo() {
    System.out.println("---Detail Information---");
```

```
    System.out.println("Audio System: " + audioSystem);
    System.out.println("Number of Doors: " + numberOfDoors);
}
}
public class program2 {
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    Vehicle vehicle = null;
    while (true) {
        System.out.println("Menu");
        System.out.println("1. Two Wheeler");
        System.out.println("2. Four Wheeler");
        System.out.println("3. Exit");
        System.out.print("Enter your choice: ");
        int choice = -1;
        try {
            choice = Integer.parseInt(sc.nextLine());
        } catch(NumberFormatException e) {
            System.out.println("Invalid input! Please enter a number.");
            continue;
        }
        if (choice == 3) {
            System.out.println("Exiting the program.");
            break;
        }

        System.out.print("Vehicle make: ");
        String make = sc.nextLine();
        System.out.print("Vehicle number: ");
        String vehicleNumber = sc.nextLine();
        System.out.print("Fuel type: ");
        String fuelType = sc.nextLine();
        System.out.print("Fuel capacity: ");
        Integer fuelCapacity;
        try {
            fuelCapacity = Integer.parseInt(sc.nextLine());
        } catch (NumberFormatException e) {
            System.out.println("Invalid input for fuel capacity. Please enter an integer.");
            continue;
        }
        System.out.print("CC: ");
        Integer cc;
        try {
            cc = Integer.parseInt(sc.nextLine());
        } catch (NumberFormatException e) {
            System.out.println("Invalid input for CC. Please enter an integer.");
            continue;
        }
        if (choice == 1) {
```

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```java
        System.out.print("Kick start available (yes/no): ");
        String kickStartInput = sc.nextLine().trim().toLowerCase();
        Boolean kickStartAvailable;
        if (kickStartInput.equals("yes")) {
            kickStartAvailable = true;
        } else if (kickStartInput.equals("no")) {
            kickStartAvailable = false;
        } else {
            System.out.println("Invalid input for kick start availability. Please enter yes or no.");
            continue;
        }
        vehicle = new TwoWheeler(make, vehicleNumber, fuelType, fuelCapacity, cc, kickStartAvailable);
    } else if (choice == 2) {
        // FourWheeler specific inputs
        System.out.print("Audio system: ");
        String audioSystem = sc.nextLine();
        System.out.print("Number of doors: ");
        Integer numberOfDoors;
        try {
            numberOfDoors = Integer.parseInt(sc.nextLine());
        } catch (NumberFormatException e) {
            System.out.println("Invalid input for number of doors. Please enter an integer.");
            continue;
        }
        vehicle = new FourWheeler(make, vehicleNumber, fuelType, fuelCapacity, cc, audioSystem,
numberOfDoors);
    } else {
        System.out.println("Invalid choice. Please choose 1, 2 or 3.");
        continue;
    }

    vehicle.displayMake();
    vehicle.displayBasicInfo();
    vehicle.displayDetailInfo();
    System.out.println();
  }
  sc.close();
}
}
```

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```
<terminated> program2 [Java Application] C:\U
Menu
1. Two Wheeler
2. Four Wheeler
3. Exit
Enter your choice: 1
Vehicle make: Ferrari
Vehicle number: TN 41 BD 0130
Fuel type: Petrol
Fuel capacity: 10
CC: 150
Kick start available (yes/no): yes
***Ferrari***
---Basic Information---
Vehicle Number: TN 41 BD 0130
Fuel Type: Petrol
Fuel Capacity: 10
CC: 150
---Detail Information---
Kick Start Available: Yes

Menu
1. Two Wheeler
2. Four Wheeler
3. Exit
Enter your choice: 2
Vehicle make: Honda
Vehicle number: TN 41 BD 0131
Fuel type: Petrol
Fuel capacity: 30
CC: 1500
Audio system: IMAX
Number of doors: 2
***Honda***
---Basic Information---
Vehicle Number: TN 41 BD 0131
Fuel Type: Petrol
Fuel Capacity: 30
CC: 1500
---Detail Information---
Audio System: IMAX
Number of Doors: 2
```

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Program 3:

```
package com.training.ooc;
class Shape {
public double calculateArea() {
    return 0.0;
}
}
class Square extends Shape {
private double side;
public Square(double side) {
    this.side = side;
}
public double getSide() { return side; }
public void setSide(double side) { this.side = side; }
@Override
public double calculateArea() {
    return side * side;
}
}
class Triangle extends Shape {
private double base;
private double height;
public Triangle(double base, double height) {
    this.base = base;
    this.height = height;
}
public double getBase() { return base; }
public void setBase(double base) { this.base = base; }
public double getHeight() { return height; }
public void setHeight(double height) { this.height = height; }
@Override
public double calculateArea() {
    return 0.5 * base * height;
}
}
class Rectangle extends Shape {
private double length;
private double breadth;
public Rectangle(double length, double breadth) {
    this.length = length;
    this.breadth = breadth;
}
public double getLength() { return length; }
public void setLength(double length) { this.length = length; }
public double getBreadth() { return breadth; }
public void setBreadth(double breadth) { this.breadth = breadth; }
@Override
```

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```
public double calculateArea() {
    return length * breadth;
}
}
public class program3 {
public static void main(String[] args) {
    Shape s;
    s = new Square(5);
    System.out.println("Area of Square: " + s.calculateArea());
    s = new Triangle(10, 7);
    System.out.println("Area of Triangle: " + s.calculateArea());
    s = new Rectangle(8, 4);
    System.out.println("Area of Rectangle: " + s.calculateArea());
}
}
```

```
<terminated> program3 [Java Appli
Area of Square: 25.0
Area of Triangle: 35.0
Area of Rectangle: 32.0
```

Program4:

```
package com.training.ooc;
import java.util.Scanner;
class Associate {
   private int associateId;
   private String associateName;
   private String workStatus;
   public Associate() {
   }
   public Associate(int associateId, String associateName, String workStatus) {
       this.associateId = associateId;
       this.associateName = associateName;
       this.workStatus = workStatus;
   }
   public int getAssociateId() {
       return associateId;
   }
   public void setAssociateId(int associateId) {
       this.associateId = associateId;
   }
   public String getAssociateName() {
       return associateName;
   }
```

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```
  public void setAssociateName(String associateName) {
      this.associateName = associateName;
  }
  public String getWorkStatus() {
      return workStatus;
  }
  private void setWorkStatus(String workStatus) {
      this.workStatus = workStatus;
  }
  public void trackAssociateStatus(int days) {
      if (days < 0) {
          setWorkStatus("Invalid days");
      } else if (days <= 20) {
          setWorkStatus("Core skills");
      } else if (days <= 40) {
          setWorkStatus("Advanced modules");
      } else if (days <= 60) {
          setWorkStatus("Project phase");
      } else {
          setWorkStatus("Deployed in project");
      }
  }
  public void displayDetails() {
      System.out.println("Associate ID: " + associateId);
      System.out.println("Associate Name: " + associateName);
      System.out.println("Work Status: " + workStatus);
  }
}
public class program4 {
  public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      Associate associate = new Associate();
      System.out.print("Enter Associate ID: ");
      int id = Integer.parseInt(sc.nextLine());
      associate.setAssociateId(id);
      System.out.print("Enter Associate Name: ");
      String name = sc.nextLine();
      associate.setAssociateName(name);
      System.out.print("Enter number of days completed in training: ");
      int days = Integer.parseInt(sc.nextLine());
      associate.trackAssociateStatus(days);
      System.out.println("\nAssociate Details:");
      associate.displayDetails();
      sc.close();
  }
}
```

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```
<terminated> program4 [Java Application] C:\Users\surya.p\.p2\p
Enter Associate ID: 10480
Enter Associate Name: Surya
Enter number of days completed in training: 20

Associate Details:
Associate ID: 10480
Associate Name: Surya
Work Status: Core skills
```